

Decentralized Coordination of Multiple Buildings With Renewable Energy Resource and Electric Vehicles

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Abstract—With the popularity of electric vehicles (EVs) and renewable energy sources (RES), the flexibility of charging and discharging of EVs and the intermittency of RES have brought challenges to building operations. Considering the mobility of EVs as commuting tools between buildings and the uncertainty of RES, it is of great practical significance to coordinate multiple buildings with RES and EVs on the premise of meeting the state of energy (SOE) requirements of the future trip. We formulate this coordination problem as a stochastic centralized mixed integer linear programming problem. A polyhedral convex set is constructed to describe the SOE uncertainty of EVs. New nonanticipative constraints (NCs) are derived through forward recursion based on constructed scenarios to guarantee the all-scenario-feasibility (ASF) and nonanticipativity of the decision. A Lagrangian relaxation-based decentralized all-scenario-feasible (LR-DASF) algorithm is developed to solve the centralized optimization problem in a decomposition and coordination way. In this method, the optimal ASF solution can be obtained with a fast convergence rate by updating Lagrangian multipliers without solving all subproblems with NCs. The performance of the LR-DASF algorithm is verified by numerical results, which shows that the algorithm can guarantee the ASF of the solution, as well as promote computational efficiency.

Note to Practitioners—EVs as energy storage devices bring energy exchanges between buildings accompanying the mobility of EVs which is an opportunity to improve the energy efficiency of multiple buildings. However, as commuting tools, the SOE of EVs must be guaranteed to be larger than the trip requirement over the randomness of RES generation. Furthermore, solving the coordinated optimization problem of multiple buildings with RES and EVs still faces computational complexity challenge due to the spatio-temporal coupling between EVs and buildings,

which may lead to costly computational effort in the premise of guaranteeing the feasibility and nonanticipativity of the decision over the uncertainties in practice. Therefore, in order to overcome the above challenges, an LR-DASF algorithm is developed in this paper to solve the coordinated optimization problem of multiple buildings with RES and EVs. Based on the algorithm, for the system operator, it updates and broadcasts the Lagrangian multipliers information to the local coordinators of buildings. For each building, the local coordinator can make ASF decisions based on new NCs with the information obtained from the system operator independently to guarantee the SOE requirement. The method developed in this paper can make faster optimal decisions without perceivable degradation in accuracy and guarantee the SOE requirement of EVs simultaneously, to meet the requirements of feasibility and computational efficiency of decision making in practice. It is conducive to the future application of LR-DASF in the coordinated optimization of buildings and EVs at the city or regional scale.

Index Terms—Decentralized optimization, renewable energy sources, all scenarios feasible method, Lagrange relaxation, coordination of EVs and buildings.

NOMENCLATURE

Indices and Sets

k	Index of time period, $k \in \{1, 2, \dots, K\}$.
l	Index of iteration, $l \in \mathbb{N}$.
m, h	Index of buildings, $m, h \in \{1, 2, \dots, M\}$.
n	Index of EVs, $n \in \{1, 2, \dots, N\}$.
s	Index of scenarios, $s \in \{1, 2, \dots, S\}$.

Parameters

P	Big number.
τ	Period duration, (h).
E^c	Battery capacity of each EV, (kWh).
$E^{\min}$	Minimum battery capacity of each EV, (kWh).
μ_0	Electric drive efficiency, (kWh/km).
ω_s	Weighting factor of scenario s .
$\xi_{m,k}$	SOE variation of EVs of building m at k , (kWh).
$p_{m,k}^{PV}$	Power from PV system of building m at k , (kW).

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$D_{n,k}^{ev}$	Location of n -th EV at k , $D_{n,k}^{ev} \in \{0, 1, 2, \dots, M\}$, 0 means on the road, others are the buildings' number.	$p_{m,k}^{ev,c}(p_{m,k}^{ev,d})$	Charging (Discharging) power of EVs of building m at k , (kW).
$d_{n,k}^{ev}$	New trip distance of n -th EV at k , (km).	$E_{m,k}^{min}(E_{m,k}^{max})$	Minimum (Maximum) SOE of EVs of building m at k , (kWh).
$E_{0,k}$	SOE of EVs on road at k , (kWh).	$z_{m,k}^{buy}(z_{m,k}^{sell})$	Binary variable. 1 means buying (selling) action; 0 otherwise.
E_k^{consd}	SOE consumed by all EVs at k , (kWh).	$z_{m,k}^{ev,c}(z_{m,k}^{ev,d})$	Binary variable. 1 means charging (discharging) action; 0 otherwise.
$E_{n,k}^{ev,need}$	SOE requirement for the new trip of n -th EV at k , (kW).	$\mathbf{a}_{m,k}$	Decision variable vector of building m at k .
$N_{m,k}$	Number of EVs of building m at k .	$\mathbf{a}_m^l$	Vector $\mathbf{a}_m$ of l -th iteration, while $\mathbf{a}_m = [\mathbf{a}_{m,1}, \dots, \mathbf{a}_{m,K}]$.
$N_{m,k}^l$	Number of EVs comes to building m during k .	$\lambda_k^{s,+}, \lambda_k^{s,-}$	Lagrange multipliers for scenario s at k .
$N_{m,k}^O$	Number of EVs departs building m during k .	$\lambda^{+,l}, \lambda^{-,l}$	Vector λ^+, λ^- of l -th iteration, while $\lambda^+ = [\lambda_k^{s,+}]_{K \times S}$, $\lambda^- = [\lambda_k^{s,-}]_{K \times S}$.
$p_{m,k}^{b,base}$	Baseline load of building m at k , (kW).		
$T_{m,k}^{ref}$	Reference thermostat setpoint temperature of building m at k , ($^{\circ}$).		
$f(T_{m,k})$	Load flexibility ratio related to thermostat setpoint temperature, $T_{m,k}$.		
$\Delta_{m,k}^-(\Delta_{m,k}^+)$	Lower (Upper) physical SOE bound of EVs of building m at k , (kWh).		
$\eta^+(\eta^-)$	Charging (Discharging) efficiency coefficient of each EV.		
$\Omega_k(\Phi_k)$	Polyhedral convex set of $\xi_{m,k}$ ($p_{m,k}^{PV}$).		
$c_k^{buy}(c_k^{sell})$	Buying (Selling) price from (to) grid at k , (RMB/kWh).		
$p^c(p^d)$	Rated charging (discharging) power of each EV, (kW).		
$p_m^{pile,c}(p_m^{pile,d})$	Rated charging (discharging) power of charging piles of building m , (kW).		
$\bar{p}_{m,k}^{ev,c}(\bar{p}_{m,k}^{ev,d})$	Maximum charging (discharging) power of EVs of building m at k , (kW).		
$\underline{p}_{m,k}^b(\bar{p}_{m,k}^b)$	Lower (Upper) building power of building m at k , (kW).		
$\underline{\xi}_{m,k}(\bar{\xi}_{m,k})$	Lower (Upper) bound of $\xi_{m,k}$, (kWh).		
$\underline{\xi}_{m,k}^s(\bar{\xi}_{m,k}^s)$	Minimum (Maximum) of $\xi_{m,k}$ in constructed scenarios, (kWh).		
$\underline{E}_{0,k}(\bar{E}_{0,k})$	Lower (Upper) bound of $E_{0,k}$, (kWh).		
$\underline{E}_{m,k}(\bar{E}_{m,k})$	Lower (Upper) bound of $E_{m,k}$, (kWh).		
$\underline{p}^G(\bar{p}^G)$	Lower (Upper) power bound of transmission line, (kW).		
$\gamma_{m,k}^1, \dots, \gamma_{m,k}^6$	Parameters1 for function $f(T_{m,k})$.		
$\kappa_{m,k}^1, \dots, \kappa_{m,k}^6$	Parameters2 for function $f(T_{m,k})$.		
$E(\cdot)$	Expectation functions.		
$g_1(\cdot), g_2(\cdot), h(\cdot)$	Ancillary functions.		
$I(\cdot)$	Indicator function.		

Variables

$E_{m,k}$	SOE of EVs of building m at k , (kWh).
$T_{m,k}$	Current thermostat setpoint temperature of building m at k , ($^{\circ}$).
$p_{m,k}^b$	Current load of building m at k , (kW).
$p_{m,k}^{buy}(p_{m,k}^{sell})$	Buying (Selling) electrical power from (to) grid of building m at k , (kW).

I. INTRODUCTION

ENERGY consumption of buildings, which accounts for 40% of the total energy consumption [1], has drawn great interests in its energy efficiency improvement for reducing carbon emissions [2], [3]. The majority of current works achieve this goal by imposing renewable energy sources (RES) and implementing energy management of integrated RES, energy storage system (ESS), and smart demand energy devices [4], [5], [6], [7]. However, the inherent intermittency of RES has a significant impact on energy efficiency due to insufficient flexibility in building energy demand.

EVs can be considered as distributed ESS when parking at buildings. On one hand, EVs show extensive flexibility of charging as the deferrable loads through grid-to-vehicle (G2V) technology [8]. On the other hand, EVs can be treated as power suppliers by discharging through vehicle-to-grid (V2G) technologies [9]. After decades of rapid growth, the quantity of global EVs has exceeded 26 million in 2022, and half of them are in China [10]. The popularity of EVs has significantly reached a prominent level that is conducive to compensating the intermittency and randomness of RES [11], [12]. Therefore, it is promising to utilize EVs as ESS to improve the energy efficiency of building energy systems.

However, as commuting tools, EVs travel between buildings, for example, leaving home to go to office buildings or other types of buildings in the morning, and returning home from office buildings or other types of buildings in the evening. Therefore, it is necessary to guarantee that the state of energy (SOE) of EVs is larger than the energy consumed on the road when leaving, which is a hard constraint. In addition, the movement of EVs between buildings not only brings energy migration but also spatio-temporal coupled constraints.

In this context, considering the uncertainty of RES generation and the spatio-temporal coupled SOE uncertainty of EVs, exploring the coordinated optimization of buildings with RES and EVs on the premise of meeting SOE requirements poses great practical difficulties:

First, nonanticipativity. It means that the decision made at the current stage should not rely on the future realization of the uncertainty [13]. So, new nonanticipative constraints (NCs) should be derived to guarantee the nonanticipativity. *Second*, all-scenario-feasibility (ASF). It means the decision

made at the current stage should deal with any possible future realization of the uncertainty to satisfy the SOE requirements of future trips. *Third*, the curse of dimensionality. The decision space grows exponentially with the rise of the scale of the buildings and EVs due to the spatio-temporal coupled characteristics.

Based on the aforementioned discussion, the motivation of this paper is to establish a coordinated formulation to address the above concerns. In comparison to existing studies, the main contributions of this paper can be summarized below.

1) The coordinated optimization of buildings with RES and EVs is formulated as a stochastic centralized mixed integer linear programming (MILP) problem considering the spatio-temporal coupled constraints and uncertainties. In this problem, the SOE of EVs is spatio-temporal coupled and uncertain, and the RES generation is also uncertain. The polyhedral convex set is employed to depict the uncertainty based on the information of EVs' travel trajectories. Then, based on the characteristics of the polyhedral convex set, any realization of the uncertainty can be represented as the convex combination of the vertices of the polyhedral convex set. Therefore, the finite scenarios constructed based on vertices of the polyhedral convex set are the prerequisite and foundation for guaranteeing the nonanticipativity and ASF in this paper.

2) New NCs are derived through forward recursion based on constructed scenarios, auxiliary variables and functions. The new NCs with auxiliary variables and functions make the decision independent of any realization of the uncertainties which means it can guarantee the nonanticipativity of the decision. Therefore, the combination of new NCs and constructed scenarios can not only guarantee the nonanticipativity, but also guarantee the ASF.

3) A Lagrangian relaxation-based decentralized all-scenario-feasible (LR-DASF) algorithm is proposed to solve the centralized optimization problem in a decomposition and coordination way. The LR-DASF algorithm makes each building focus on its subproblem through the local coordinator according to the Lagrangian multipliers from the system operator, and the system operator updates new Lagrangian multipliers without collecting all decisions as long as any building can reduce the Lagrangian function. The ASF of the optimal solution is also proved.

The performance of the LR-DASF algorithm is verified by a series of case studies with parameters acquired from government websites or actual data. Firstly, a small-scale deterministic case study is tested to validate the energy efficiency improvement of the method developed in this paper. Then, another small-scale case study under uncertainty is tested to validate the ASF. Finally, large-scale case studies are tested to demonstrate computational efficiency improvement, accuracy and scalability. Numerical results show that the solution of LR-DASF is less than 0.01% compared with the centralized method, while the computational efficiency is improved by at least 6 times. The results shed light on the coordination optimization of multiple buildings with the increasing number of EVs in practical engineering applications. Our method can make faster optimal decisions without perceivable degradation in accuracy and guarantee the

SOE requirement of EVs simultaneously. It is conducive to the future application of LR-DASF in the coordinated optimization of buildings and EVs at the city or regional scale.

This paper is organized as follows. In Section II, the related works are briefly reviewed. In Section III, the problem is formulated as a stochastic MILP problem under uncertainty. In Section IV, the LR-DASF algorithm is presented. In Section V, case studies and numerical results are shown. In Section VI, the limitations of our work and future works are discussed. A brief conclusion is summarized in Section VII.

II. LITERATURE REVIEW

In recent years, more and more attentions have been attracted to the coordinated optimization of buildings integrating RES and EVs due to the flexibility of charging and discharging during the long parking time at the building.

In terms of the scale of buildings, they are mainly classified as coordinated optimization of a single building or multiple buildings. For a single building, EVs are treated as ESS or deferrable loads. For example, Sangswang and Konghirun [14] developed and implemented optimal control strategies on a hardware platform for a home microgrid integrating PV, ESS, and EVs. Saber et al. [15] developed a novel transactive energy market scheme for a residential building to determine the charging/discharging action of plug-in EVs and then employed the model predictive control (MPC) algorithm to tackle the uncertainties of demand and PV generation. Liu et al. [16] proposed a transactive real-time EV charging management scheme for a commercial building to track the fluctuating PV generation. Wu et al. [17] employed a two-stage stochastic programming framework for a large office building to coordinate EVs and PV, optimizing the energy management of the building. Xu et al. [18] proposed a scenario-based stochastic optimization method for a shopping mall considering PV and EVs.

For the coordination of multiple buildings, EVs are regarded as portable energy storage media. For example, Ojand and Dagdougui [19] presented a curtailable demand response scheduling model in a residential community exploiting the flexibility of RES and EVs. Du et al. [20] presented a coordinated energy dispatched scheme for autonomous microgrids equipped with RES, ESS and Plug-in EVs. Yan et al. [21] presented the optimized operation of an EV charging station and a commercial building aiming to reduce operational costs. It can be seen that the above works have improved the energy efficiency of buildings by the coordination of RES and EVs. However, those works ignore the spatio-temporal coupled flexibility obtained by the mobility of EVs between buildings.

Recently, some researchers have focused on the coordination of multiple buildings considering the spatio-temporal coupled flexibility caused by EVs' mobility. For example, Roccotelli et al. [22] adopted the MonteCarlo simulation to generate scenarios for the smart district energy management of buildings, EVs, RES and ESS, representing the mobility of EV and other uncertainties. Huang et al. [23] and Yang et al. [24] both employed Markov chain to depict the mobility of EV.

An event-based control policy is proposed for the coordination of buildings equipped with PV, EV charging and hydrogen energy storage [23]. A distributed policy is proposed for the coordination of EV charging and wind power in a microgrid of buildings [24]. However, these works have exploited the flexibility of individual EVs rather than the aggregated EVs, which makes them unsuitable for large-scale EVs because the decision space exponentially increases with the rise of EVs number. Therefore, this paper focuses on the coordination of flexibility of aggregated EVs, and multiple buildings with RES, considering the uncertainties of RES generation and SOE of EVs, on the premise of meeting SOE requirements for the next trip.

The optimization problem with uncertainties has been studied for decades, and there are many methods to tackle the uncertainties. Those methods can be divided into two groups: the first group is based on the distribution information, such as the scenario-based method [22]. It can achieve better expected objective value but may be infeasible for the scenarios not included. The other group is based on the uncertainty set [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], such as the two-stage robust optimization (RO) methods [25], [31], [32], [33], the adjustable RO methods [27], [28], [29], [34], the multistage RO methods [35], the distributed RO methods [27], [28], and the all-scenario-feasible (ASF) method [13], [36]. The two-stage RO methods are to find a certain solution for the worst scenario, and the optimal solutions tend to be rather conservative and infeasible in other certain situations. The adjustable RO methods and the multistage RO methods are developed to address the conservativeness, and the decision process is usually much more complicated because the decision cannot be made until the realization of the uncertainty happens. The distributed RO methods are developed to improve the computational efficiency under Lagrangian relaxation, while the conservativeness and the infeasibility are the same as the two-stage RO methods. The ASF methods are introduced to address the nonanticipativity and infeasibility. Due to the characteristics of EVs, as commuting tools, the feasibility of decision-making is the most important thing, which can guarantee the SOE requirements for the trip. So the ASF method is adopted in this paper to guarantee that the decision made at the current stage can deal with any possible future realization of the uncertainties to satisfy the SOE requirements of future trips.

Due to the curse of dimensionality caused by the spatio-temporal coupled constraints, the computational burden of the ASF method increases exponentially with the rise of the scale of the buildings and EVs. Decentralization is an effective technique to improve computational efficiency and scalability [24], [27], [28], [37], compared to centralized approaches. One form of the decentralized framework, as shown in [37], solves all subproblems in parallel without the system operator exchanging information, however, its convergence and efficiency can not be guaranteed due to the spatio-temporal coupled flexibility caused by EVs' mobility. The second form of the decentralized framework, as shown in [24], [27], and [28], also solves all subproblems in

parallel, but with the system operator exchanging information with different buildings, which may have better performance in efficiency with the spatio-temporal coupled uncertainty. Therefore, this paper proposes a decentralized ASF algorithm under the Lagrangian relax framework, i.e. LR-DASF, to solve the coordinated optimization problem in a decomposition and coordination way to guarantee nonanticipativity, ASF and computational efficiency.

For convenience, a detailed comparison between our work and existing works is summarized in Table I. The comparison mainly takes 7 aspects into account including the terms of building number, the spatio-temporal coupled uncertainty, nonanticipativity, ASF, etc.

III. PROBLEM FORMULATION

In this section, the problem of the coordination of multiple buildings with RES and EVs under uncertainties is formulated with the objective of minimizing the system operation cost, on the premise of meeting the SOE requirements for the coming trip. The system description is shown in subsection III-A. The model of EVs and the model of building load's flexibility are presented in subsections III-B and III-C, respectively. The operation constraints and the objective function of the problem are developed in subsections III-D and III-E, respectively.

A. System Description

A microgrid of buildings and EVs is shown in Fig.1. It consists of M buildings and N EVs. Buildings comprise one or several types, e.g. residential, office, commercial, etc. Each building is equipped with solar PV panels on the roof and charging piles in the underground parking lots. EVs can travel on the road or park at the buildings.

It is assumed that: 1) the installation and maintenance fees of PV panels are ignored since we consider the operation problem in this paper, as similar to [23]. 2) EVs have bidirectional power exchange capability through charging piles: G2V and V2G [8]. 3) The building load affected by thermostat setpoint temperature can only be shifted other than curtailed due to the comfort requirements of occupants [38]. 4) The buying (selling) price from (to) grid is time-of-use (TOU) pricing [7]. 5) Each building has a local coordinator and the whole microgrid has only one system operator. The local coordinators can get the global information (e.g. Lagrange multiplier) from the system operator and then make decisions for the subproblem. The system operator can spread the global information to the local coordinators.

The discrete version of this problem is formulated and the scheduling horizon is discretized into K periods. The time of one period can be determined based on practice requirements, without affecting operation details and computational efficiency, e.g., it can be 1 hour or 30 minutes. The objective function is to minimize the total operation cost on the condition that all constraints are satisfied.

B. Model of EVs

As similar to [39], in this paper, the Markov chain is adopted to depict the mobility of a single EV whereas the

TABLE I
THE COMPARISON BETWEEN OUR WORK AND EXISTING WORKS

Reference	Flexibility of aggregated EVs	Multiple buildings	ESS or EVs	Spatio-temporal coupled uncertainty	Nonanticipativity	ASF of solution	Algorithm
[14], [17], [21]	×	×	both	×	×	×	Scenario-based
[15], [16], [18]	×	×	EVs	×	×	×	Scenario-based
[19], [20]	×	✓	both	×	×	×	MPC
[22]	×	✓	both	✓	×	×	Scenario-based
[23]	×	✓	both	✓	×	×	Event-based
[24]	×	✓	EVs	✓	×	×	Event-based
Our work	✓	✓	EVs	✓	✓	✓	LR-DASF

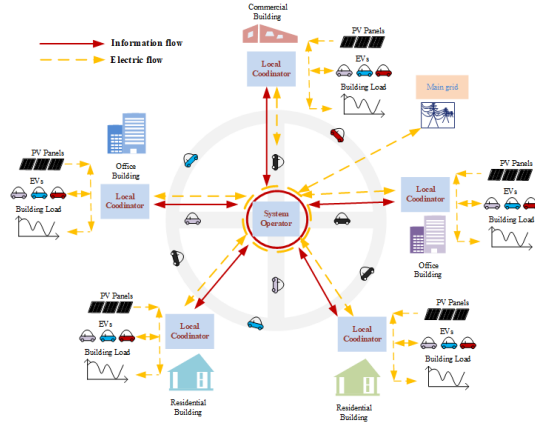


Fig. 1. The system structure of multiple buildings.

state is the location of EV, $D_{n,k}^{ev}$. If EV n starts its new trip during period k , the trip length, $d_{n,k}^{ev}$, is modeled as truncated power exponential distribution, and the corresponding SOE requirement is calculated by (1).

$$E_{n,k}^{ev_need} = \mu_0 \cdot d_{n,k}^{ev}. \quad (1)$$

Equation (1) means the SOE requirement of the trip is proportional to the distance [40].

In this paper, the destination, parking time and distance of the trip are predicted through historical data. For each building, as EVs are treated as a whole ESS, then the related information of the aggregated EVs is calculated, such as the number of EVs at/ out/ to building m during period k ($N_{m,k}, N_{m,k}^O, N_{m,k}^I$) [39]. Those information can affect the charging /discharging power of EVs which is shown in the latter constraints.

$$I(A) = \begin{cases} 0, & \text{if event } A \text{ is false} \\ 1, & \text{if event } A \text{ is true,} \end{cases} \quad (2)$$

$$N_{m,k} = \sum_{n=1}^N I(D_{n,k} = m), \quad (3)$$

$$N_{m,k}^O = \sum_{n=1}^N I(D_{n,k} = m \ \& \ D_{n,k+1} \neq m), \quad (4)$$

$$N_{m,k}^I = \sum_{n=1}^N I(D_{n,k} \neq m \ \& \ D_{n,k+1} = m), \quad (5)$$

$$N_{m,k+1} = N_{m,k} + N_{m,k}^I - N_{m,k}^O. \quad (6)$$

Equation (2) gives the definition of the indicator function, which is employed to calculate the numbers of aggregated EVs of each building. Equations (3) - (5) show the details of the calculation of numbers of EVs at/ out/ to building m during period k . Equation (6) describes the EV quantity evolution of each building.

C. Model of Building Load's Flexibility

Since this paper focuses on the distributed coordination of multiple buildings with RES and EVs, similar to [39], we use a simplified engineering model to formulate the flexibility of the building loads. Note that other physical models can also be applied to the formulation of the flexibility of the building loads and have no impact on the contribution of this paper. Therefore, the flexibility of the building load is formulated as a piecewise linear function of thermostat setpoint temperature as shown below.

$$p_{m,k}^b = p_{m,k}^{b_base} \cdot (1 + f(T_{m,k})), \quad (7)$$

$$f(T_{m,k}) = \begin{cases} \gamma_{m,k}^1 + \kappa_{m,k}^1 \cdot \Delta T_{m,k}, & \Delta T_{m,k} \in [-3, -2] \\ \gamma_{m,k}^2 + \kappa_{m,k}^2 \cdot \Delta T_{m,k}, & \Delta T_{m,k} \in [-2, -1] \\ \gamma_{m,k}^3 + \kappa_{m,k}^3 \cdot \Delta T_{m,k}, & \Delta T_{m,k} \in [-1, 0] \\ \gamma_{m,k}^4 + \kappa_{m,k}^4 \cdot \Delta T_{m,k}, & \Delta T_{m,k} \in [0, 1] \\ \gamma_{m,k}^5 + \kappa_{m,k}^5 \cdot \Delta T_{m,k}, & \Delta T_{m,k} \in [1, 2] \\ \gamma_{m,k}^6 + \kappa_{m,k}^6 \cdot \Delta T_{m,k}, & \Delta T_{m,k} \in [2, 3], \end{cases}$$

$$\Delta T_{m,k} = T_{m,k} - T_{m,k}^{ref}, \quad (8)$$

where $p_{m,k}^b$ and $p_{m,k}^{b_base}$ are the current building load after temperature adjustment and the baseline building load with the reference temperature of building m during period k , respectively. $T_{m,k}$ and $T_{m,k}^{ref}$ are the current thermostat setpoint temperature and the reference setpoint temperature of building m during period k , respectively. $\Delta T_{m,k}$ is the temperature difference of building m during period k between the current setpoint temperature and the reference setpoint temperature of thermostat. $f(T_{m,k})$ is the ratio of building m during period k decided by the current thermostat setpoint temperature. Parameters, $\gamma_{m,k}^1, \dots, \gamma_{m,k}^6$ and $\Delta_{m,k}^1, \dots, \Delta_{m,k}^6$, can be obtained from the historical data of building load, outside air temperature and the temperature difference of thermostat, are the offset and slope rate of building m during period k for the piecewise linear function.

Generally speaking, those parameters are different for different buildings. Even for the same building, they may be different in time, depending on whether the dynamics of the building loads during weekdays and weekends are different or not. For example, $p_{m,k}^{b_base}$ in (7) is decided by the predicted outside weather factors, the reference setpoint temperature of thermostat and the historical data. If the historical data is quite different between weekdays and weekends, $p_{m,k}^{b_base}$ for weekdays and weekends are different, and vice versa [41].

Equation (8) shows that the building load's flexibility is affected by the temperature difference between the current temperature and the reference temperature of thermostat. For example, when the temperature is set higher than the reference value in summer, $\Delta T_{m,k}$ and $f(T_{m,k})$ are negative, which means building load will be lower than the baseline load, and vice versa.

D. Operation Constraints

1) Power Balance Constraints:

$$p_{m,k}^{buy} - p_{m,k}^{sell} + p_{m,k}^{PV} = p_{m,k}^b + p_{m,k}^{ev_c} - p_{m,k}^{ev_d}, \quad (9)$$

$$E_{m,k+1} = E_{m,k} + (\eta^+ p_{m,k}^{ev_c} - \frac{p_{m,k}^{ev_d}}{\eta^-}) \cdot \tau - \xi_{m,k}. \quad (10)$$

Equation (9) describes power balance of building m at period k . It shows buildings can purchase (sell) electricity from (to) the power grid when necessary, and EVs parked in building m at period k are treated as a whole ESS. Equation (10) describes SOE dynamics of EVs parked at building m between adjacent periods $k, k+1$. Equations (9) - (10) show that EVs can charge or discharge like a whole ESS.

2) Constraints of Building Loads:

$$\sum_{k=1}^K p_{m,k}^b = \sum_{k=1}^K p_{m,k}^{b_base}, \quad (11)$$

$$\underline{T}_{m,k} \leq T_{m,k} \leq \bar{T}_{m,k}, \quad (12)$$

$$\underline{p}_{m,k}^b \leq p_{m,k}^b \leq \bar{p}_{m,k}^b. \quad (13)$$

Equation (11) imposes that the total consumption over the schedule horizon is the same before and after the temperature adjustment of the thermostat. According to the category of devices, the thermostat belongs to load shifting device, which means that the building load affected by the thermostat is shifted in time rather than curtailed [42]. Equation (12) guarantees thermostat setpoint temperature is in the comfort zone. Equation (13) implies the constraint of building load capacity.

3) Constraints of EVs:

$$\underline{E}_{m,k} \leq E_{m,k} \leq \bar{E}_{m,k}, \quad (14)$$

$$\xi_{m,k} \in [\underline{\xi}_{m,k}, \bar{\xi}_{m,k}], \quad (15)$$

$$(E_{m,k} - \xi_{m,k}) \in [\Delta_{m,k}^-, \Delta_{m,k}^+]. \quad (16)$$

Equation (14) means the SOE of EVs must be larger than the SOE requirement for the next trip ($\underline{E}_{m,k}$) as commuting

tools. At the same time, it is not exceed battery capacity ($\bar{E}_{m,k}$) which is physical constraint. The value range of stochastic variable, $\xi_{m,k}$, is defined in (15). The SOE of remaining EVs after mobility should also be satisfied as (16).

4) Constraints of Charging and Discharging Power of EVs:

$$0 \leq p_{m,k}^{ev_c} \leq z_{m,k}^{ev_c} \cdot \bar{p}_{m,k}^{ev_c}, \bar{p}_{m,k}^{ev_c} = \min\{N_{m,k} \cdot p^c, p_m^{pile_c}\}, \quad (17)$$

$$0 \leq p_{m,k}^{ev_d} \leq z_{m,k}^{ev_d} \cdot \bar{p}_{m,k}^{ev_d}, \bar{p}_{m,k}^{ev_d} = \min\{N_{m,k} \cdot p^d, p_m^{pile_d}\}, \quad (18)$$

$$z_{m,k}^{ev_c} + z_{m,k}^{ev_d} \leq 1. \quad (19)$$

Equations (17) and (18) show that the maximum charging and discharging power are decided by the number of EVs at the building and the charging piles' capacity. Equation (19) implies charging and discharging actions cannot be both happened at the same time.

5) Constraints of Power Exchange Between Microgrid and Main Grid:

$$\underline{p}^G \leq \sum_{m=1}^M (p_{m,k}^{buy} - p_{m,k}^{sell}) \leq \bar{p}^G, \quad (20)$$

$$0 \leq p_{m,k}^{buy} \leq z_{m,k}^{buy} \cdot P, \quad (21)$$

$$0 \leq p_{m,k}^{sell} \leq z_{m,k}^{sell} \cdot P, \quad (22)$$

$$z_{m,k}^{buy} + z_{m,k}^{sell} \leq 1. \quad (23)$$

The bidirectional power exchange range between the microgrid and main grid is limited due to the transmission line constraint as shown in (20). The constraints (21) - (23) imply that the purchasing or selling power of each building is also limited and they cannot both happen at the same time.

6) Constraints of Stochastic Variables:

$$\xi_k \in \Omega_k, \quad (24)$$

$$\mathbf{p}_k^{PV} \in \Phi_k, \quad (25)$$

where $\xi_k = [\xi_{1,k}, \xi_{2,k}, \dots, \xi_{M,k}]^T$ is a stochastic variable vector representing the uncertainty of SOE change of EVs. And $\mathbf{p}_k^{PV} = [p_{1,k}^{PV}, p_{2,k}^{PV}, \dots, p_{M,k}^{PV}]^T$ is a stochastic variable vector representing the uncertainty of PV power. Ω_k, Φ_k in (24) and (25) are polyhedral convex sets which will be introduced in next subsection IV-A

E. The Objective Function

$$(P1) \min J, J = \sum_{k=1}^K \sum_{m=1}^M (c_k^{buy} \cdot p_{m,k}^{buy} - c_k^{sell} \cdot p_{m,k}^{sell}) \cdot \tau. \quad (26)$$

As all buildings are the price-taker, the objective is to minimize the overall operation cost subjected to the constraints of (6) - (25).

IV. SOLUTION METHODOLOGY

The stochastic MILP problem (P1) cannot be directly solved due to the uncertainty of the infinite realizations. Secondly, it is costly to solve due to the curse of dimensionality because the decision space grows exponentially with the building scale.

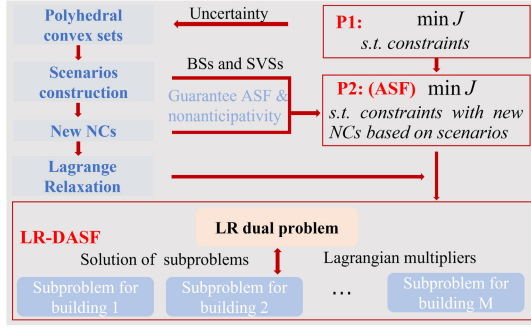


Fig. 2. The flowchart of the proposed algorithm.

To overcome these challenges, an LR-DASF algorithm is developed to solve the problem. The charflow is shown in Fig.2.

Firstly, polyhedral convex sets are constructed to capture the uncertainties in subsection IV-A. Then, base scenarios (BSs) and selected vertex scenarios (SVSs) are constructed based on the vertices of uncertainty sets in subsection IV-B. Thirdly, new NCs based on constructed scenarios are proposed to guarantee the ASF and nonanticipativity of the solution in subsection IV-C. And, the ASF-based formulation with NCs is proposed in subsection IV-D. Finally, to improve the computational efficiency, the LR-DASF algorithm is developed to solve the problem in a decomposition and coordination way in subsection IV-E.

A. Uncertainty Sets

As mentioned above, there exist two stochastic variables in problem (P1), $\xi_k, \mathbf{p}_k^{PV}$. The general method is to employ the probability information to depict them. However, the truth is the probability information is usually unknown in many circumstances. Therefore, we will adopt another method to depict those stochastic variables, the uncertainty sets.

For the convenience of description, those two stochastic variables are combined into a new vector, denoted as $\varphi_k = [p_k^{PV}, \xi_k]$. In this section, we will use two different polyhedral convex sets to model them according to their different stochastic characteristics.

As known, for ξ_k , it has spatio-temporal coupled characteristics caused by the mobility between buildings and the hard constraints of SOE requirements for the next trip. Based on the principle of energy conservation, (27) can be obtained to describe SOE conservation of the whole EVs between adjacent periods, which is spatio-temporal coupled.

$$E_{0,k+1} + \sum_{m=1}^M E_{m,k+1} = E_{0,k} + \sum_{m=1}^M E_{m,k} + \sum_{m=1}^M \left(\eta^+ p_{m,k}^{ev-c} - \frac{p_{m,k}^{ev-d}}{\eta^-} \right) \cdot \tau - E_k^{consd}. \quad (27)$$

Then, a new expression of SOE conservation about stochastic variable ξ_k is shown in (28), which is derived by

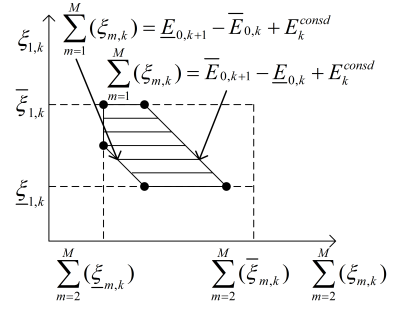


Fig. 3. The projection of the uncertainty set on 2-dimensions.

subtracting (10) from (27) on both sides of the equation.

$$E_{0,k+1} = E_{0,k} + \sum_{m=1}^M (\xi_{m,k}) - E_k^{consd}. \quad (28)$$

Therefore, an M -dimensional polyhedral convex set is constructed as (29) to capture the uncertainty of ξ_k based on (28) and the SOE boundary of EVs, which is shown in Fig.3.

$$\Omega_k = \left\{ \xi_k \left| \begin{array}{l} \xi_{m,k} \leq \bar{\xi}_{m,k}; m = 1, \dots, M \\ E_{0,k} \leq E_{0,k} \leq \bar{E}_{0,k}; \\ E_{0,k+1} \leq E_{0,k} + \sum_{m=1}^M (\xi_{m,k}) - E_k^{consd} \leq \bar{E}_{0,k+1} \end{array} \right. \right\}. \quad (29)$$

For $\mathbf{p}_k^{PV}$, a box set defined in (30) is employed to capture the uncertainty of PV power since the PV power generation is bounded by the weather conditions.

$$\Phi_k = \{ \mathbf{p}_k^{PV} | p_{m,k}^{PV} \leq p_{m,k}^{PV} \leq \bar{p}_{m,k}^{PV}; m = 1, \dots, M \}. \quad (30)$$

B. Scenarios Construction

Since the uncertainty sets have been constructed in the above section, the scenarios constructed in this section based on the uncertainty sets are different from the scenarios generated by the traditional method, such as Monte Carlo (MC) simulation. The constructed scenarios in this section are divided into two types: base scenarios (BSs) and selected vertex scenarios (SVSs). BSs are constructed to guarantee economic performance, and SVSs are constructed to guarantee the nonanticipativity and ASF.

The SVSs are intended to be constructed of the vertices of the uncertainty sets. The uncertainty sets are the polyhedral convex sets, any value in those sets can be represented as a convex combination of the vertices. In other words, any realization of the uncertainties can also be a convex combination of the SVSs. It means that if solutions of the SVSs exist, there always exists a solution for the realization, which is ASF. This is the cornerstone of transforming infinite realizations of the uncertainties into the convex combination of finite SVSs. The total number of SVSs is close to 2^{2M} , far less than the original scale of 2^{2MK} .

BSs are usually selected as the significant scenarios because they are considered as “the most probable scenarios”, such as the expected value of the uncertainties or scenarios generated

from historical data. The number of BSs can be 0,1 or any natural number according to practical requirements. BSs can also be generated by MC simulation. The number of the BSs can only affect the economic performance and computational efficiency. Too many BSs will bring a heavy computational burden rather than economic performance improvement. In this paper, only one BS is considered which is the expected scenario to get better economic performance with less computational burden.

The detailed construction methods are introduced below.

1) BSs are constructed as the expected value of φ_k in this paper, similar to [13] in this paper.

$$BSs = \{s | \mathbf{p}_k^{PV,s} = E(\mathbf{p}_k^{PV}), \xi_k^s = E(\xi_k)\}. \quad (31)$$

2) SVSs are constructed into three types, $SVSs = SVS1 \cup SVS2 \cup SVS3$, which is different from [13] in which SVSs are constructed in the direction of increasing the total sum of variables in all periods. Due to the temporal coupled constraint in (29), scenarios with maximum /minimum sum value in all periods cannot be constructed. The detailed construction process is shown in (32) - (37).

$$SVS1 = \{s | \sum_{m=1}^M \varphi_{m,k}^s = \underline{Q}_k, \sum_{m=1}^M \varphi_{m,k+1}^s = \overline{Q}_{k+1}\}, \quad (32)$$

$$SVS2 = \{s | \sum_{m=1}^M \varphi_{m,k}^s = \overline{Q}_k, \sum_{m=1}^M \varphi_{m,k+1}^s = \underline{Q}_{k+1}\}, \quad (33)$$

$$SVS3 = \{s | \sum_{m=1}^M \varphi_{m,k}^s \in (\underline{Q}_k, \overline{Q}_k), \sum_{m=1}^M \varphi_{m,k+1}^s \in (\underline{Q}_{k+1}, \overline{Q}_{k+1})\}, \quad (34)$$

$$Q_k(E_{0,k+1}, E_{0,k}, \mathbf{p}_k^{PV}) = E_{0,k+1} - E_{0,k} + E_k^{consd} + \sum_{m=1}^M p_{m,k+1}^{PV}, \quad (35)$$

$$\underline{Q}_k = E_{0,k+1} - \overline{E}_{0,k} + E_k^{consd} + \sum_{m=1}^M p_{m,k+1}^{PV}, \quad (36)$$

$$\overline{Q}_k = \overline{E}_{0,k+1} - E_{0,k} + E_k^{consd} + \sum_{m=1}^M \overline{p}_{m,k+1}^{PV}. \quad (37)$$

Equation (32) shows SVS1 is a scenario set with maximum sum value at odd periods and minimum sum value at even periods. Equation (33) shows SVS2 is a scenario set with maximum sum value at even periods and minimum sum value at odd periods. Equation (34) shows SVS3 is constructed following the same method as [13].

We will use Fig.4 as an example to illustrate the construction of SVSs. Period k is the odd period. The numbers of vertices at period $k, k+1$ are different. Then, virtual vertices are created as any one of the existing vertices to make the numbers the same, virtual vertex v_k^1 is created as vertex v_k^5 here. SVSs are given below.

SVS1 are given as $\{(v_k^2, v_{k+1}^5), (v_k^3, v_{k+1}^4)\}$.

SVS2 are given as $\{(v_k^4, v_{k+1}^3), (v_k^5, v_{k+1}^2)\}$.

SVS3 are given as $\{(v_k^1, v_{k+1}^1)\}$.

C. Nonanticipative Constraints

In this subsection, new NCs are derived based on scenarios constructed by vertices of the uncertainty sets. Those NCs

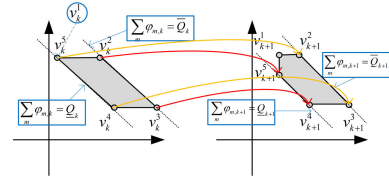


Fig. 4. Illustration of SVSs construction.

do not rely on any realization of the uncertainty, which can guarantee nonanticipativity. Meanwhile, those NCs make decisions at the current period feasible to any future realization of the uncertainty, which is ASF. The details are shown below.

Firstly, three ancillary functions, $h(x)$, $g_1(E_{m,k}, \xi_{m,k})$, $g_2(E_{m,k}, \xi_{m,k})$, are defined as (38) - (40). Then two ancillary parameters are introduced, $E_{m,k}^{\max}, E_{m,k}^{\min}$, which are the maximum and minimum value of the SOE of EVs parked at building m at period k . They can determine a new compact feasible region of $E_{m,k}$ by (41), which can guarantee that if $E_{m,k}$ is in the feasible region, there always exists a feasible solution no matter what realization of the uncertainty will be.

$$h(x) = \begin{cases} \frac{1}{\eta^+ \tau} x, & \text{if } (x \geq 0) \\ \frac{1}{\eta^- \tau} x, & \text{if } (x \leq 0), \end{cases} \quad (38)$$

$$g_1(E_{m,k}, \xi_{m,k}) = \min(\Delta_{m,k}^+, E_{m,k} - \xi_{m,k}), \quad (39)$$

$$g_2(E_{m,k}, \xi_{m,k}) = \max(\Delta_{m,k}^-, E_{m,k} - \xi_{m,k}), \quad (40)$$

$$E_{m,k} \leq E_{m,k}^{\min} \leq E_{m,k} \leq E_{m,k}^{\max} \leq \overline{E}_{m,k}. \quad (41)$$

Secondly, based on two ancillary functions (39) - (40), ancillary parameters and constructed scenarios, as well as the upper and lower bounds of constraints (14), (15) and (16), the new constraints of remaining EVs after departure can be obtained as (44).

$$\underline{\xi}_{m,k} \leq \xi_{m,k}^s; \xi_{m,k}^s = \min_s(\xi_{m,k}^s), s \in SVSs, \quad (42)$$

$$\overline{\xi}_{m,k}^s \leq \overline{\xi}_{m,k}; \overline{\xi}_{m,k}^s = \max_s(\xi_{m,k}^s), s \in SVSs, \quad (43)$$

$$g_2(E_{m,k}^{\min}, \overline{\xi}_{m,k}^s) \leq E_{m,k} - \xi_{m,k} \leq g_1(E_{m,k}^{\max}, \underline{\xi}_{m,k}^s). \quad (44)$$

Finally, new NCs are derived as (45) - (48) from constraints (9) - (11), (13), (17) - (23) and (44) through forward recursion based on constructed scenarios, ancillary functions and variables. New NCs ((45) - (46)) are constraints for adjacent periods, and (47) - (48) are constraints for the whole periods. It can be seen that new NCs make the decision dependent on auxiliary variables and functions rather than any realization of the uncertainties, which means it can guarantee the nonanticipativity of the decision. Furthermore, New NCs can guarantee ASF, which means if SOE is in the feasible range decided by $E_{m,k}^{\max}, E_{m,k}^{\min}$, no matter what future realization of the uncertainties will be, it always has a solution.

$$h(E_{m,k+1}^{\max} - g_2(E_{m,k}^{\min}, \overline{\xi}_{m,k}^s)) \leq \min(\overline{p}_{m,k}^{ev-c}, P + \overline{p}_{m,k}^{PV} - \underline{p}_{m,k}^b), \quad (45)$$

$$h(E_{m,k+1}^{\min} - g_1(E_{m,k}^{\max}, \underline{\xi}_{m,k}^s)) \geq \max(-\overline{p}_{m,k}^{ev-d}, -P + \underline{p}_{m,k}^{PV} - \overline{p}_{m,k}^b), \quad (46)$$

$$\sum_{k=1}^K \sum_{m=1}^M h \left(E_{m,k+1}^{\max} - g_2 \left(E_{m,k}^{\min}, \bar{\xi}_{m,k}^s \right) \right) \leq K \cdot \bar{p}^G + \sum_{k=1}^K \sum_{m=1}^M (\bar{p}_{m,k}^{PV} - p_{m,k}^{b_base}), \quad (47)$$

$$\sum_{k=1}^K \sum_{m=1}^M h \left(E_{m,k+1}^{\min} - g_1 \left(E_{m,k}^{\max}, \underline{\xi}_{m,k}^s \right) \right) \geq K \cdot \underline{p}^G + \sum_{k=1}^K \sum_{m=1}^M (p_{m,k}^{PV} - p_{m,k}^{b_base}). \quad (48)$$

D. The ASF Formulation With New NCs

Based on constructed scenarios, ancillary parameters and functions, and new NCs, the original problem (P1) can be converted into the problem (P2).

$$(P2) \min J, J = \sum_{k=1}^K \sum_{m=1}^M \sum_{s=1}^S \omega^s \cdot (c_k^{buy} \cdot p_{m,k}^{buy,s} - c_k^{sell} \cdot p_{m,k}^{sell,s}) \cdot \tau, \quad (49)$$

subject to constraints (7) - (23), (41) - (48) based on constructed scenarios (BSs and SVSs). All decision variables of building m in period k form a vector $\mathbf{a}_{m,k}$. $\mathbf{a}_{m,k} = [p_{m,k}^{buy,s}, p_{m,k}^{sell,s}, p_{m,k}^{ev_c,s}, p_{m,k}^{ev_d,s}, z_{m,k}^{buy,s}, z_{m,k}^{sell,s}, z_{m,k}^{ev_c,s}, z_{m,k}^{ev_d,s}]$,

$E_{m,k}^s, E_{m,k}^{\max}, E_{m,k}^{\min}, p_{m,k}^b, T_{m,k}]$.

Theorem 1: If $(p_{m,k}^b, T_{m,k}, E_{m,k}^{\max}, E_{m,k}^{\min})_{M \times K}$ is feasible for the problem (P2), then it is all-scenario-feasible for PV generation and SOE variation.

The proof process of *Theorem 1* is shown in Appendix.

According to Theorem 1, the solution to the problem (P2) can guarantee nonanticipativity without using any future realization of the uncertainties but the constructed scenarios (BSs and SVSs). Furthermore, the solution is feasible for any future realization of the uncertainties under SOE requirements to guarantee ASF.

E. LR-DASF Algorithm

As mentioned above, an ASF solution can be obtained by solving the problem (P2) in a centralized form, however, it is costly to solve as the number of scenarios is close to 2^{2M} . Therefore, a Lagrangian relaxation-based decentralized all-scenario-feasible (LR-DASF) algorithm is proposed.

The LR-DASF algorithm solves the centralized optimization problem in a decomposition and coordination way. In the LR-DASF algorithm, each building focuses on its subproblem through the local coordinator according to the Lagrangian multipliers from the system operator, and the system operator updates new Lagrangian multipliers without collecting all decisions, as long as any building can reduce the Lagrangian function. The detail is introduced below.

Firstly, introduce Lagrangian multipliers $\{\lambda_k^{s,+}, \lambda_k^{s,-}, k = 1, 2, \dots, K; s \in BSs, SVSs\}$ to decouple the constraint (20).

Algorithm 1 LR-DASF Algorithm

Step1. Initialize. Set $l = 0$, l is the iteration index. Given initial multipliers λ^0 .

$$\lambda^0 = (\lambda_k^{s,+}, \lambda_k^{s,-}), k \in \{1, 2, \dots, K\}, s \in \{1, 2, \dots, S\}.$$

Step2. Update the Lagrangian multipliers

for $m = 1 : M$ **do**

Solve subproblem (51) with λ^l to obtain $L_m^{*,l}$ and $\mathbf{a}_m^l$.

if $l \geq 1$ **then**

If condition (53) is satisfied, update decision variables

$$\mathbf{a}_{h:h \neq m}^l : \mathbf{a}_{h:h \neq m}^l = \mathbf{a}_{h:h \neq m}^{l-1};$$

end if

Break;

end for

Update new directions for Lagrangian multipliers.

$$g_{\lambda_k^{s,+},l}^l = \omega^s \cdot \left(\sum_{m=1}^M (p_{m,k}^{buy,s,l} - p_{m,k}^{sell,s,l}) - \bar{p}^G \right).$$

$$g_{\lambda_k^{s,-},l}^l = \omega^s \cdot \left(\underline{p}^G - \sum_{m=1}^M (p_{m,k}^{buy,s,l} - p_{m,k}^{sell,s,l}) \right).$$

Choose step size s^l which must satisfy below condition.

$$0 < s^l < (L^* - L^l) / \|\mathbf{g}^l\|^2,$$

where L^* is the optimal value.

$$L^l = L(\lambda^l) = \sum_{m=1}^M L_m^{*,l} - \sum_{k=1}^K \sum_{s=1}^S \omega^s \cdot [\lambda_k^{s,+},l \cdot \bar{p}^G - \lambda_k^{s,-},l \cdot \underline{p}^G]$$

$$\mathbf{g}^l = (g_{\lambda_1^{1,+},l}^l, \dots, g_{\lambda_K^{1,+},l}^l, \dots, g_{\lambda_1^{S,+},l}^l, \dots, g_{\lambda_K^{S,+},l}^l, g_{\lambda_1^{1,-},l}^l, \dots, g_{\lambda_K^{1,-},l}^l, \dots, g_{\lambda_1^{S,-},l}^l, \dots, g_{\lambda_K^{S,-},l}^l).$$

Update the multipliers according to

$$\lambda^{s,-,l+1} = \lambda^{s,-,l} + s^l \cdot g_{\lambda_k^{s,-},l}^l; \lambda^{s,+,l+1} = \lambda^{s,+,l} + s^l \cdot g_{\lambda_k^{s,+},l}^l.$$

Step3. Check stop criteria. If the following criterion is met, go to step 4; otherwise update $l = l + 1$ and go to step 2.

$$\|\lambda^{l+1} - \lambda^l\| < \varepsilon,$$

where ε is the accuracy requirement.

Step4. Construct feasible solution. Based on solutions of subproblems and dual problem, construct a feasible solution of the original problem.

Then the LR function, L , can be derived as (50):

$$\begin{aligned} \min L, L = & \sum_{k=1}^K \sum_{m=1}^M \sum_{s=1}^S \omega^s \cdot (c_k^{buy} \cdot p_{m,k}^{buy,s} - c_k^{sell} \cdot p_{m,k}^{sell,s}) \cdot \tau \\ & + \sum_{k=1}^K \sum_{s=1}^S \omega^s \cdot [\lambda_k^{s,-} \cdot (\underline{p}^G - \sum_{m=1}^M (p_{m,k}^{buy,s} - p_{m,k}^{sell,s})) \\ & + \lambda_k^{s,+} \cdot (\sum_{m=1}^M (p_{m,k}^{buy,s} - p_{m,k}^{sell,s}) - \bar{p}^G)], \end{aligned} \quad (50)$$

subject to constraints (7) - (19), (21) - (23), (41) - (48) based on constructed scenarios (i.e., BSs and SVSs).

Secondly, construct a separative optimization subproblem for each building. For m -th building, it is denoted as L_m by collecting all items related to building m , which is expressed

as (51).

$$\min L_m, L_m = \sum_{k=1}^K \sum_{s=1}^S \omega^s \cdot [(c_k^{buy} \cdot \tau - \lambda_k^{s,-} + \lambda_k^{s,+}) \cdot p_{m,k}^{buy,s}] - (c_k^{sell} \cdot \tau - \lambda_k^{s,-} + \lambda_k^{s,+}) \cdot p_{m,k}^{sell,s}, \quad (51)$$

subject to constraints (7) - (19), (21) - (23), (41) - (48) based on selected scenarios (i.e., BSs and SVSs).

Thirdly, update multipliers by solving the dual problem for the system operator.

$$\max_{\lambda_k^{s,+}, \lambda_k^{s,-} \geq 0} q, q = \sum_{m=1}^M L_m^* - \sum_{k=1}^K \sum_{s=1}^S \omega^s \cdot [\lambda_k^{s,+} \cdot \bar{p}^G - \lambda_k^{s,-} \cdot \underline{p}^G]. \quad (52)$$

To achieve a fast convergence rate, the surrogate subgradient method is adopted in our algorithm. A proper search direction for convergence is obtained without optimally solving all subproblems if the decision variable vector for building m at l -th iteration, $\mathbf{a}_m^l$, can reduce the Lagrangian function L , satisfying the following surrogate optimization condition [43].

$$L(\lambda^{s,+l}, \lambda^{s,-l}, a_m^l, a_{h:h \neq m}^{l-1}) < L(\lambda^{s,+l}, \lambda^{s,-l}, a_m^{l-1}, a_{h:h \neq m}^{l-1}). \quad (53)$$

The detailed algorithm is introduced in Algorithm 1.

V. CASE STUDIES AND NUMERICAL RESULTS

In this section, the performance of computational efficiency, accuracy and scalability of the LR-DASF method is tested through three case studies. General parameters of case studies are shown in subsection V-A. A small-scale deterministic case study is tested to validate the model and algorithm in subsection V-B. A small-scale case study under uncertainty is tested to validate the ASF in subsection V-C. Finally, large-scale case studies are tested in subsection V-D to demonstrate the computational efficiency, accuracy and scalability of the LR-DASF algorithm. All tests are implemented by using Matlab R2019b and Gurobi 9.0.2 at Intel(R) Core (TM) i7-8700T CPU@ 2.40 GHz PC with 16 GB RAM.

A. Description of Case Studies

The LR-DASF method is tested through three case studies. The scheduling horizon is 24 hours which is divided into 24 periods, so the time length of each period is 1 hour. According to the Beijing transportation report [44], most people leave home between 7:00 am and 9:00 am and get off work between 5:00 pm and 7:00 pm. Then, the probability matrices are generated from the above data, which are the same as [39]. The EVs are simplified as the same brand, Nissan leaf, and the related parameters are listed in Table II [45]. The electricity price is adopted as the time of use price of Beijing obtained from the website of the Beijing Municipal Commission of Development and Reform [46], which is listed in Table III. Based on the daily routine of humans, the thermostats in buildings are closed during time 22:00 - 05:00 and open during time 06:00 - 21:00 to do load shifting according to [38]. The hourly solar radiation density

TABLE II
BATTERY SPECIFICATION OF EV

Parameter	Setting	Parameter	Setting
E^c	40 kWh	$p^{ev-c}(p^{ev-d})$	17.6 kW
$\eta^+(\eta^-)$	0.9	μ_0	0.184 kWh/km

TABLE III
THE ELECTRICITY PRICE (RMB)

Period	c_k^{buy}	c_k^{sell}	Period	c_k^{buy}	c_k^{sell}
23-06	0.3113	0.3	11-12 16	1.4397	0.3
13-14 18-20 10	1.3104	0.3	7-9 15 17 21-22	0.7847	0.3

is generated from the website of the U.S. Department of Energy based on typical summer weather data of Beijing [47]. Different scales of buildings and EVs in Beijing are tested as examples in the section.

Firstly, a small-scale deterministic case study of 5 buildings (Building 1 and 2 are residential buildings, Building 3 and 4 are office buildings, and Building 5 is a commercial building) and 300 EVs is tested to validate the model and algorithm in subsection V-B. PV panels for each building are 500, 600, 2000, 2500, and 3000 m^2 , respectively. The charging/discharging power capacities of buildings are 2000, 2000, 4000, 4000, and 4000 kW, respectively.

Secondly, as compared with the traditional methods, a small-scale case study with the same configuration as the first case study under uncertainty is tested to validate the ASF in subsection V-C.

Finally, large-scale case studies are tested in subsection V-D to demonstrate the computational efficiency, accuracy and scalability of the algorithm developed in this paper. The number of buildings in the studies has increased from 5 to 10, and the number of EVs has increased from 300 to 3000. We will explore whether the LR-DASF algorithm exhibits good scalability performance, and whether the key factor affecting the computational efficiency and accuracy is the number of buildings or the number of EVs.

B. Performance Analysis of Deterministic Problem

In this subsection, the LR-DASF method is tested to validate the model and algorithm. The coordination results are exhibited in Table IV, Fig. 5 and Fig. 6. The reason why only the results of Building 1, 3, and 5 (representing residential, office, and commercial buildings) are shown is that the optimization results of the same type of buildings are similar.

Table IV shows the operation cost of each building before and after coordination. It is found that, through coordination, the operation cost of each building after coordination can be reduced by at least 10% compared with the individual optimization.

In Fig. 5, it shows EVs, together with PV, have improved the energy efficiency of the building, making the power exchange between the building and the main grid close to

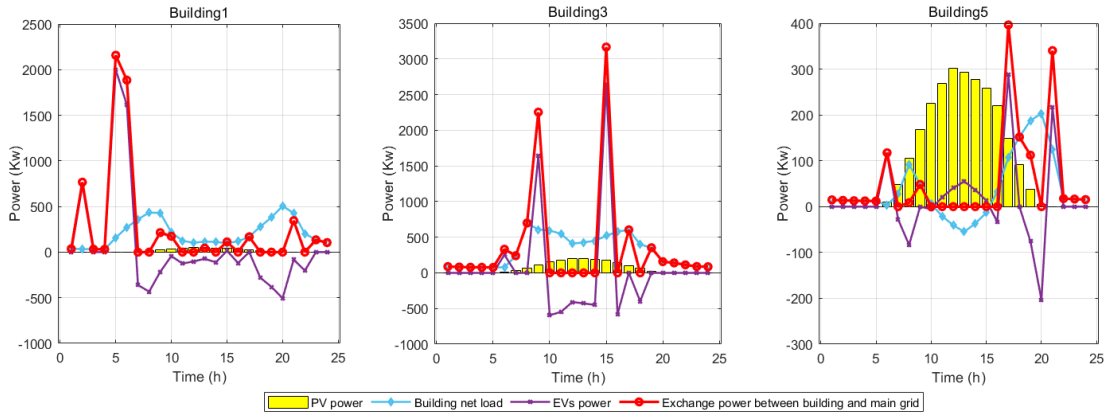


Fig. 5. Coordinative optimization result of buildings.

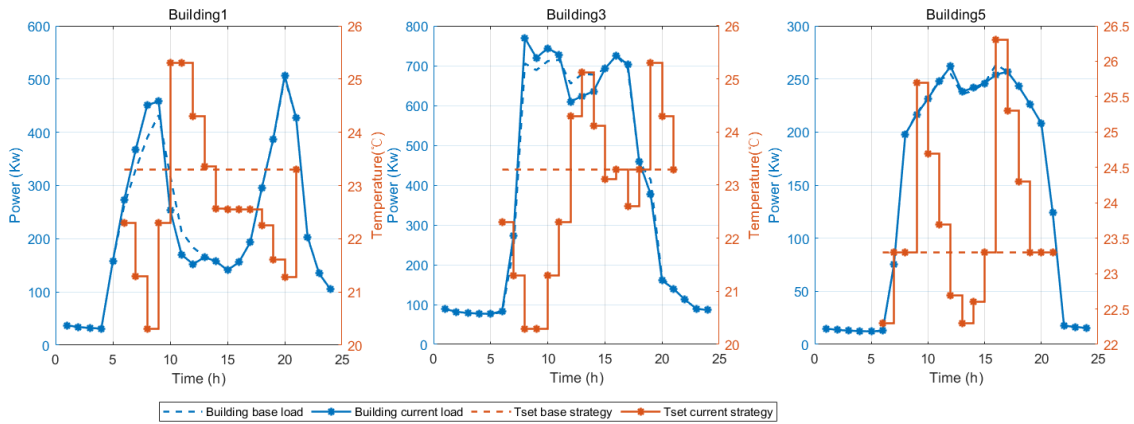


Fig. 6. Thermostat setpoint temperature strategy for buildings.

TABLE IV
OPERATION COST SAVING OF EACH BUILDING

	Building1	Building2	Building3	Building4	Building5
Cost of coordination ($\times 10^3$ RMB)	2.547	2.4704	6.6116	6.2596	1.05
Cost of individual optimization ($\times 10^3$ RMB)	5.4092	5.2585	8.0297	7.4915	1.217
Cost saving	52.91%	53.02%	17.66%	16.44%	13.72%

0 as much as possible. For an individual building, EVs can either store surplus energy from the PV system when PV power is sufficient or purchase energy from the grid when the electricity price is low, and feed energy back to the grid or building when the electricity price is high. For example, for Building 5 (commercial building), EVs charge to store energy during periods 10-14 when PV power is surplus and discharge during periods 7-8 and 19-20 when the electricity price is high. The power exchange between Building 5 and the main grid during periods 10-15 is correspondingly approaching 0. The same situation happens in Building 3 and Building 1. For the whole microgrid, EVs can transfer energy between buildings. For example, EVs store energy during periods of lower electricity prices when parking at Building 1, and then feed back to Building 3 and Building 5, which corresponds to

the process of EVs leaving residential buildings and going to the workplace in the morning. EVs store energy free or at a lower electricity price when parking at Building 3 and Building 5 either from the PV system or the grid, and then feed back to Building 1, which corresponds to the process of EVs returning home from the workplace in the evening.

In Fig. 6, it shows building loads are shifted from high price periods to low price periods by the thermostat setpoint temperature strategies which are the precooling strategies. Set a lower temperature in the morning to store cooling by the building thermal mass when the electricity price is low to pre-cool the building. Then set a higher temperature in the noon with high electricity price to reduce energy consumption and cost.

C. ASF Analysis of the Problem Under Uncertainty

In this subsection, the ASF of the proposed LR-DASF method is analyzed by comparing it with scenario-based stochastic programming (SSP) [22], [39]. Fig. 7 shows SOE regions decided by LR-DASF and SSP, and Table V gives detailed information of period 9 of Building 4. These two results are combined to explain why the LR-DASF method can guarantee the ASF of the solution.

TABLE V
SOE REGION INFORMATION

Time	SOE range (LR-DASF)	SOE range (SSP)	$\xi_{4,k}$	$\bar{p}_{m,k}^{ev,c}$ ($\bar{p}_{m,k}^{ev,d}$)
9	[1584,4320]	[1408,4320]	[-988.21,-216]	4000
10	[1579.77,5400]	[3410.27,5400]	—	—

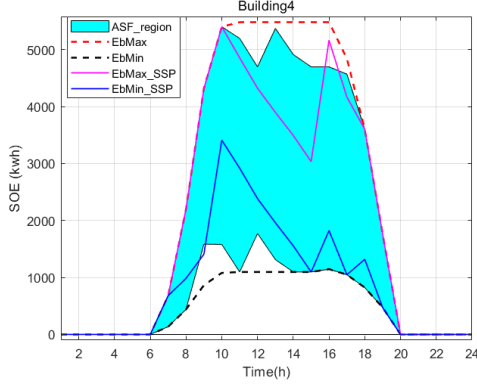


Fig. 7. Feasible SOE range comparison of EVs.

TABLE VI
SOLUTIONS OF LR-DASF AND SSP FOR EXPERIMENT 1

Case No.	$\xi_{4,9}$	SSP		LR-DASF	
		$p_{4,9}^{ev,c}$	$p_{4,9}^{ev,d}$	$p_{4,9}^{ev,c}$	$p_{4,9}^{ev,d}$
1	-988.21	3337.54	0	3141.99	0
2	-600	3768.89	0	3573.33	0
3	-400	3991.11	0	3795.56	0
4	-300	4102.22	0	3906.67	0
5	-216	4195.56	0	4000	0

In Fig. 7, it shows SOE regions of LR-DASF and SSP are both more compact than the original region, but it cannot tell why LR-DASF can guarantee the ASF but SSP can't. Without loss of generality, we take period 9 of Building 4 as an example to explain the reason, which is shown in Table V. ASF means if SOE is in the feasible range decided by the decision variables, $(E_{m,k}^{\max}, E_{m,k}^{\min})_{M \times K}$, no matter what future realization of the uncertainty, $\xi_{4,9}$, will be, it always has a solution.

Table V shows that the differences in SOE regions of $E_{4,9}, E_{4,10}$ between LR-DASF and SSP are the lower bounds. Then we will just take the lower bounds as examples to explicitly explain the ASF. As known, the worst scenario must be one of the vertices of the polyhedral convex set. For $\xi_{4,9}$, the worst scenario must be $\xi_{4,9} = -216$ or $\xi_{4,9} = -988.21$. So both of them are included for the verification.

Then, two experiments are designed to verify the ASF of LR-DASF and SSP.

Experiment 1: fix the variables $E_{4,9} = 1408, E_{4,10} = 5400$ for SSP, and $E_{4,9} = 1584, E_{4,10} = 5400$ for LR-DASF. Then verify the feasibility of the solution for different realizations of $\xi_{4,9}$, the results are shown in Table VI.

TABLE VII
SOLUTIONS OF LR-DASF AND SSP FOR EXPERIMENT 2

Case No.	$\xi_{4,9}$	SSP		LR-DASF	
		$p_{4,9}^{ev,c}$	$p_{4,9}^{ev,d}$	$p_{4,9}^{ev,c}$	$p_{4,9}^{ev,d}$
1	-988.21	0	1708.39	0	3355.60
2	-600	0	1359	0	3006.21
3	-400	0	1179	0	2826.21
4	-300	0	1089	0	2736.21
5	-216	0	1013.4	0	2660.61

Table VI shows, for SSP, the solutions of No.4 and No.5 highlighted in red, $p_{4,9}^{ev,c}$, are larger than the maximum $\bar{p}_{4,9}^{ev,c} = 4000$, which means they are infeasible and can not guarantee the ASF. For LR-DASF, the solutions of these five cases including the worst case are all feasible. Furthermore, if we represent the uncertainty $\xi_{4,9}$ as $\xi_{4,9} = -988.21 + \pi \cdot (-216 + 988.21)$, $\pi \in [0, 1]$, then $p_{4,9}^{ev,c} = 3141.99 + \pi \cdot (4000 - 3141.99)$ is feasible for any realization of $\xi_{4,9}$ depending on different values of π . It can be verified that LR-DASF can guarantee the ASF.

Experiment 2: fix the variables $E_{4,9} = 4320, E_{4,10} = 3410$ for SSP, and $E_{4,9} = 4320, E_{4,10} = 1579.77$ for LR-DASF. Then verify the feasibility of the solution for different realizations of $\xi_{4,9}$, the results are shown in Table VII.

Table VII shows, the solutions for all cases are feasible for SSP and LR-DASF. It can also be verified that they can guarantee the ASF in this experiment.

In summary, SSP can not guarantee the ASF because there exist infeasible cases, but LR-DASF can. Following the proof of Theorem 1 shown in Appendix, it can be verified that the range determined by LR-DASF, no matter what value the stochastic variable $\xi_{4,9}$ will be, $p_{m,k}^{ev,c}$ and $p_{m,k}^{ev,d}$ always exist, which means ASF.

D. Performance Analysis of the LR-DASF Algorithm

In this subsection, the computational efficiency, economic performance and scalability of the proposed LR-DASF algorithm are validated through large-scale case studies, compared with traditional algorithms, such as the centralized ASF algorithm (CASF), and the nested column and constraints generation algorithm (Nested CCG) for two-stage robust optimization with integer recourse [32]. The comparison results of computational time and economic performance are listed in Table VIII and Table IX, respectively.

It is found in Table VIII that LR-DASF can improve the computational time at least 6 times as compared with the other two methods. The larger the problem scale, the better the performance of the LR-DASF algorithm. Moreover, the LR-DASF algorithm shows good performance in scalability especially compared with the Nested CCG. When the building number equals 10, Nested CCG has no solution, but the LR-DASF algorithm not only has a solution but also has the fastest computational efficiency. Furthermore, the computational time of LR-DASF is more sensitive to building scale rather than the scale of EVs since EVs are treated in a whole ESS way.

TABLE VIII
COMPUTATIONAL TIME OF LR-DASF, CASF AND NESTED CCG

Case No.	Buildings No.	EVs No.	CASF (s)	LR-DASF (s)	Nested CCG (s)
1	5	300	7.87	1.27	2074.12
2	5	500	7.44	1.90	2116.96
3	5	700	6.69	2.60	3103.44
4	7	700	774.35	27.21	1054.23
5	7	900	138.48	24.50	2002.09
6	7	1100	192.02	38.49	1001.01
7	10	1100	1156.96	84.59	—
8	10	2000	1068.59	239.66	—
9	10	3000	1142.38	166.12	—

TABLE IX
SYSTEM COST OF LR-DASF, CASF AND NESTED CCG

Case No.	Buildings No.	EVs No.	CASF ($\times 10^4$ RMB)	LR-DASF ($\times 10^4$ RMB)	Nested CCG ($\times 10^4$ RMB)
1	5	300	1.9193	1.9518	2.2929
2	5	500	1.9044	1.9115	2.2610
3	5	700	2.0537	2.0580	2.2849
4	7	700	3.5131	3.5167	4.3467
5	7	900	3.5128	3.5214	4.3915
6	7	1100	3.5604	3.5713	4.5137
7	10	1100	4.5514	4.5545	—
8	10	2000	4.6244	4.6529	—
9	10	3000	5.5176	5.5195	—

It is found in Table IX that CASF has the best performance whereas the Nested CCG is the worst. The gap between LR-DASF and CASF is quite small, no more than 0.01%. Therefore, the numerical results show that the LR-DASF results in a reduction of more than 80% in computation time without perceivable degradation in accuracy.

VI. DISCUSSION

In this paper, we focus on the coordinated optimization of multiple buildings with RES and EVs, aiming to reveal the flexibility brought to building energy management by the use of EVs. EVs could play a similar role to ESS when parking at buildings, as they can be charged as loads or discharged as power suppliers. On the other hand, EVs act like mobile batteries when EVs move between buildings as commuting tools, bringing energy transfer between different buildings. As commuting tools, the SOE requirements of EVs for the coming trip are the top priority and must be met. Therefore, the feasibility of the decision-making for charging, discharging and satisfying the SOE requirements for future trips is the most important for the method developed in this paper. It encourages EVs to actively participate in the coordination without affecting their travel plans under any

realization of the uncertainties, which is helpful to improve energy efficiency and RES utilization. So integrating ESS into the system has no impact on the method developed in this paper.

As pointed out in [22], the uncertainties in the coordinated optimization of multiple buildings with RES and EVs can be divided into three categories: uncertainty in the supply side, uncertainty in the demand side and uncertainty in energy pricing. In this paper, we consider two of the most common uncertainties closely related to decision feasibility while guaranteeing the nonanticipativity caused by the uncertainties. The first one is the RES generation which represents the uncertainty in the supply side, and the other one is the SOE of EVs caused by the mobility which represents the uncertainty in the demand side. Note that the uncertainty in energy pricing can also be addressed by our method with the box set, but it has no impact on the decision feasibility.

In this paper, the uncertainties are depicted by polyhedral convex sets rather than probabilistic distributions. Then scenarios are constructed based on the vertices of the uncertainty sets, and new NCs are derived based on constructed scenarios to guarantee ASF and nonanticipativity of the decision. Finally, the LR-DASF algorithm is developed to solve the centralized optimization problem in a decomposition and coordination way to promote computational efficiency and scalability. It is verified that the proposed LR-DASF algorithm can guarantee the ASF of the solution to any realization of the uncertainties with less number of scenarios and higher computational efficiency. Therefore, we hope this work brings insight and extension to the coordination of multiple buildings.

In this paper, there still exist some limitations. Firstly, all buildings are supposed as price-takers. Although the TOU energy pricing (purchasing/selling) scheme is conventionally adopted based on policy, there exists a real-time pricing rate in practice. The operation cost of our method could be decreased and the performance could be improved with the real-time pricing rate adopted in the future. In addition, the performance of ASF could also address the uncertainty of energy pricing. So it is a new perspective and viewpoint to the coordinated optimization of multiple buildings. Secondly, the battery capacity of the current system without ESS is less than that of integrating ESS. Due to the battery capacity, The performance of ASF is weaker than that of integrating ESS. With the development of ESS, the disadvantages of maintenance costs and huge space will be improved [48], [49], [50]. ESS will be widely used in buildings. Hence, the battery capacity and the performance of ASF will be greater if ESS is integrated into the system in the future. Therefore, it is an interesting future work to do cooperation of multiple buildings equipped with ESS and PV, and EVs to improve energy efficiency, RES utilization and economic performance.

VII. CONCLUSION

In this paper, we investigate the coordination of multiple buildings with PV and EVs. This problem is formulated as a centralized MILP problem, aiming to fully utilize PV by the charging/discharging of EVs to improve energy efficiency and reduce the system operation cost, on the

premise of meeting the SOE requirements of the future trip. Due to the uncertainties of SOE of EVs and PV power, the nonanticipativity and ASF of the decision cannot be guaranteed by the current constraints. New NCs are constructed based on constructed scenarios to guarantee the nonanticipativity and ASF of the decision. By analyzing the structure of this centralized form, the LR-DASF algorithm is developed to decouple the spatially coupled constraints between buildings, converting the centralized optimization problem into a decentralized form. Each building focuses on its subproblem according to the Lagrangian multipliers from the system operator, and the system operator can update new Lagrangian multipliers without collecting all decisions as long as any building can reduce the Lagrangian function. Moreover, the performance of the proposed method is demonstrated by numerical tests in deterministic and uncertain cases with different scales through comparisons with traditional algorithms (i.e., Nested CCG and SSP). The results show that the LR-DASF algorithm can guarantee nonanticipativity and ASF of the solution on the premise of meeting the hard constraints of SOE requirements. Besides, the LR-DASF algorithm results in a reduction of more than 80% in computation time without perceivable degradation in accuracy.

APPENDIX PROOF OF THEOREM 1

Proof: To prove Theorem 1, it equals to prove the following two conclusions.

1) Any solution $(p_{m,k}^b, T_{m,k}, E_{m,k}^{\max}, E_{m,k}^{\min})_{M \times K}$ is feasible to all possible realization $[\varphi_{m,k}]_{2M \times K}$.

2) Nonanticipativity of decision is satisfied by any solution to the problem (P2).

A. Proof of Conclusion 1

For convenience, variable $p_{m,k}^{ev}, p_{m,k}^g$, are introduced as (A1) - (A3). Because $h(x)$ is a monotone function, then constraints (19) and (23) are naturally guaranteed.

$$p_{m,k}^{ev} = p_{m,k}^{ev,c} - p_{m,k}^{ev,d} \quad (A1)$$

$$p_{m,k}^g = p_{m,k}^{buy} - p_{m,k}^{sell} \quad (A2)$$

$$p_{m,k}^{ev} = h(E_{m,k+1} - (E_{m,k} - \xi_{m,k})) \quad (A3)$$

According to problem (P2), each $s \in SVS$, there must exist feasible solution $[p_{m,k}^{buy,s}, p_{m,k}^{sell,s}, p_{m,k}^{ev,c,s}, p_{m,k}^{ev,d,s}, z_{m,k}^{buy,s}, z_{m,k}^{sell,s}, z_{m,k}^{ev,c,s}, z_{m,k}^{ev,d,s}, E_{m,k}^s]_{M \times K}$. All constraints are satisfied.

$$p_{m,k}^{g,s} + p_{m,k}^{PV,s} = p_{m,k}^b + p_{m,k}^{ev,s} \quad (A4)$$

$$p_{m,k}^{ev,s} = h(E_{m,k+1}^s - (E_{m,k}^s - \xi_{m,k}^s)) \quad (A5)$$

$$E_{m,k}^{\min} \leq E_{m,k}^s \leq E_{m,k}^{\max} \quad (A6)$$

$$-\bar{p}_{m,k}^{ev,d} \leq p_{m,k}^{ev,s} \leq \bar{p}_{m,k}^{ev,c} \quad (A7)$$

$$-P \leq p_{m,k}^{g,s} \leq P \quad (A8)$$

$$\underline{p}^G \leq \sum_{m=1}^M (p_{m,k}^{g,s}) \leq \bar{p}^G \quad (A9)$$

Firstly, for any realization of φ_k , it can be expressed as a convex combination of vertices $\varphi_k^s (s \in SVS)$, in other words, there exists a group of convex combination coefficients.

$$\{\beta_k^s | \beta_k^s \geq 0, \sum_{s \in SVS} \beta_k^s = 1\} \quad (A10)$$

$$\varphi_k = \sum_{s \in SVS} \beta_k^s \cdot \varphi_k^s \quad (A11)$$

$$\xi_{m,k} = \sum_{s \in SVS} \beta_k^s \cdot \xi_{m,k}^s; p_{m,k}^{PV} = \sum_{s \in SVS} \beta_k^s \cdot p_{m,k}^{PV,s} \quad (A12)$$

Secondly, define two subsets S_1, S_2 defined by (A13) and (A14).

$$S_1 = \{s | \sum_{m=1}^M \varphi_{m,k}^s = \bar{Q}_k\} \quad (A13)$$

$$S_2 = \{s | \sum_{m=1}^M \varphi_{m,k}^s = \underline{Q}_k\} \quad (A14)$$

Then, for the realization φ_k , there always exists two scenarios $s_1 \in S_1, s_2 \in S_2$, make the (A15) always right. For any value $E_{m,k} \in [E_{m,k}^{\min}, E_{m,k+1}^{\max}]$, we can get constraints (A16)-(A17).

$$\varphi_{m,k}^{s_2} \leq \varphi_{m,k} \leq \varphi_{m,k}^{s_1}; s_1 \in S_1, s_2 \in S_2 \quad (A15)$$

$$p_{m,k}^{PV} \leq p_{m,k}^{PV} \leq \bar{p}_{m,k}^{PV} \quad (A16)$$

$$E_{m,k}^{\min} - \xi_{m,k} \leq E_{m,k} - \xi_{m,k} \leq E_{m,k}^{\max} - \xi_{m,k} \quad (A17)$$

For the sake of presentation, we will use $h(g_2(\xi))$ as abbreviation of $h(E_{m,k+1}^{\max} - g_2(E_{m,k}^{\min}, \xi_{m,k}))$, $h(g_1(\xi))$ for $h(E_{m,k+1}^{\min} - g_2(E_{m,k}^{\max}, \xi_{m,k}))$. Furthermore, new constraints about the upper boundary and lower boundary in (A17) are derived as (A18) - (A21).

$$g_1(E_{m,k}^{\max}, \xi_{m,k}^{s_1}) \leq g_1(E_{m,k}^{\max}, \xi_{m,k}) \leq g_1(E_{m,k}^{\max}, \xi_{m,k}^{s_2}) \quad (A18)$$

$$g_2(E_{m,k}^{\min}, \xi_{m,k}^{s_1}) \leq g_2(E_{m,k}^{\min}, \xi_{m,k}) \leq g_2(E_{m,k}^{\min}, \xi_{m,k}^{s_2}) \quad (A19)$$

$$h(g_2(\xi_{m,k}^{s_2})) \leq h(g_2(\xi_{m,k})) \leq h(g_2(\xi_{m,k}^{s_1})) \quad (A20)$$

$$h(g_1(\xi_{m,k}^{s_2})) \leq h(g_1(\xi_{m,k})) \leq h(g_1(\xi_{m,k}^{s_1})) \quad (A21)$$

Thirdly, three solution sets Ψ_1, Ψ_2, Ψ_3 are defined by (A22), (A23) and (A24).

$$\Psi_1 = \left\{ p_k^{ev,s_1} \left| \begin{array}{l} p_{m,k}^{ev,s_1} \leq h(g_2(\xi_{m,k}^{s_1})) \\ p_{m,k}^{ev,s_1} \geq h(g_1(\xi_{m,k}^{s_1})) \\ h(g_2(\xi_{m,k}^{s_1})) \leq P + \bar{p}_{m,k}^{PV} - p_{m,k}^b \\ h(g_1(\xi_{m,k}^{s_1})) \geq -P + \bar{p}_{m,k}^{PV} - p_{m,k}^b \\ h(g_2(\xi_{m,k}^{s_1})) \leq \bar{p}_{m,k}^{ev,c} \\ h(g_1(\xi_{m,k}^{s_1})) \geq -\bar{p}_{m,k}^{ev,d} \\ \sum_{m=1}^M h(g_2(\xi_{m,k}^{s_1})) \leq \bar{p}^G + \sum_{m=1}^M (\bar{p}_{m,k}^{PV} - p_{m,k}^b); \\ \sum_{m=1}^M h(g_1(\xi_{m,k}^{s_1})) \geq \underline{p}^G + \sum_{m=1}^M (\bar{p}_{m,k}^{PV} - p_{m,k}^b) \\ s_1 \in S_1; \end{array} \right. \right\} \quad (A22)$$

$$\Psi_2 = \left\{ \mathbf{p}_k^{ev,s_2} \left| \begin{array}{l} p_{m,k}^{ev,s_2} \leq h(g_2(\xi_{m,k}^{s_2})) \\ p_{m,k}^{ev,s_2} \geq h(g_1(\xi_{m,k}^{s_2})) \\ h(g_2(\xi_{m,k}^{s_2})) \leq P + p_{m,k}^{PV} - p_{m,k}^b \\ h(g_1(\xi_{m,k}^{s_2})) \geq -P + p_{m,k}^{PV} - p_{m,k}^b \\ h(g_2(\xi_{m,k}^{s_2})) \leq \bar{p}_{m,k}^{ev,c} \\ h(g_1(\xi_{m,k}^{s_2})) \geq -\bar{p}_{m,k}^{ev,d} \\ \sum_{m=1}^M h(g_2(\xi_{m,k}^{s_2})) \leq \bar{p}^G + \sum_{m=1}^M (p_{m,k}^{PV} - p_{m,k}^b); \\ \sum_{m=1}^M h(g_1(\xi_{m,k}^{s_2})) \geq \underline{p}^G + \sum_{m=1}^M (p_{m,k}^{PV} - p_{m,k}^b) \\ s_2 \in S_2; \end{array} \right. \right\} \quad (A23)$$

$$\Psi_3 = \left\{ \mathbf{p}_k^{ev} \left| \begin{array}{l} p_{m,k}^{ev} \leq h(g_2(\xi_{m,k})) \\ p_{m,k}^{ev,s} \geq h(g_1(\xi_{m,k})) \\ h(g_2(\xi_{m,k})) \leq P + p_{m,k}^{PV} - p_{m,k}^b \\ h(g_1(\xi_{m,k})) \geq -P + p_{m,k}^{PV} - p_{m,k}^b \\ h(g_2(\xi_{m,k})) \leq \bar{p}_{m,k}^{ev,c} \\ h(g_1(\xi_{m,k})) \geq -\bar{p}_{m,k}^{ev,d} \\ \sum_{m=1}^M h(g_2(\xi_{m,k})) \leq \bar{p}^G + \sum_{m=1}^M (p_{m,k}^{PV} - p_{m,k}^b); \\ \sum_{m=1}^M h(g_1(\xi_{m,k})) \geq \underline{p}^G + \sum_{m=1}^M (p_{m,k}^{PV} - p_{m,k}^b) \end{array} \right. \right\} \quad (A24)$$

It can be seen that Ψ_1, Ψ_2 are not empty, because $\{p_{m,k}^{ev,s_1} \mid s_1 \in S_1\} \subseteq \Psi_1$ and $\{p_{m,k}^{ev,s_2} \mid s_2 \in S_2\} \subseteq \Psi_2$.

As known, suppose $x_1 \leq x \leq x_2$, $x_3 \leq x \leq x_4$ and $x_1 \leq x_3 \leq x_2 \leq x_4$, then $x_5 \leq x \leq x_6$ is not empty if $x_1 \leq x_5 \leq x_3$ and $x_2 \leq x_6 \leq x_4$. Then the solution set defined by (A24) is not empty.

As function $h(x)$ is monotonically increasing, then if $x_1 \leq x_2$, there must be $h(x_1) \leq h(x_2)$, and vice versa.

$$\begin{aligned} h(E_{m,k+1} - (E_{m,k} - \xi_{m,k})) &\geq h(E_{m,k+1}^{\min} - g_1(E_{m,k}^{\max}, \xi_{m,k})) \\ h(E_{m,k+1} - (E_{m,k} - \xi_{m,k})) &\leq h(E_{m,k+1}^{\max} - g_2(E_{m,k}^{\min}, \xi_{m,k})) \end{aligned} \quad (A25)$$

As known, suppose $x \in [\underline{x}, \bar{x}]$, $\underline{y} \leq x \leq \bar{y}$, $\forall x$ is valid if and only if $\underline{y} \leq \underline{x}$, $\bar{x} \leq \bar{y}$, there must exist a feasible solution according to [36]. For any value $E_{m,k} \in [E_{m,k}^{\min}, E_{m,k+1}^{\max}]$, (A25) is always established, which means there exists a feasible solution, $p_{m,k}^{ev}$.

Proof of Conclusion 2: It is found that decision variables are constructed only based on the realization of stochastic variables at period k . In other words, the future realization of the uncertainties is not used at the current period k . Therefore, the nonanticipativity of the decision is satisfied. Q.E.D.

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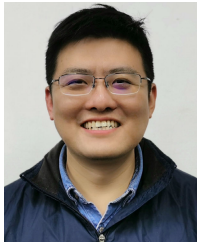
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